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Conflict of Interest and Funding Source:

A co-author of this paper, A/Prof David Opar is listed as a co-inventor on a patent, filed by the Queensland University of Technology (QUT), for a field-testing device of eccentric hamstring strength, which is now known commercially as the NordBord. A/Prof Opar has received revenue distributions from QUT based on revenue that QUT has generated through the commercialisation of his intellectual property. A/Prof Opar is a minority shareholder in

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Vald Performance Pty Ltd, the company responsible for commercialization of the NordBord and ForceDecks, among other devices. A/Prof Opar has received research funding from Vald Performance, for work unrelated to the current manuscript. A/Prof Opar was previously the Chair of the Vald Performance Research Committee, a role that was unpaid. A/Prof Opar has family members who are minor shareholders and/or employees of Vald Performance. Co-authors of this paper, Christopher Pollard and Joshua Ruddy, were previously employed with Vald Performance who manufacture the Nordbord and ForceDecks used in this study. However, neither were employed by Vald Performance at the time of completing this study.

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ABSTRACT

Purpose: Investigate impact of an isometric (ISO) or Nordic hamstring exercise (NHE) intervention, alongside a sprint training program on hamstring strength, architecture and sprinting performance in Australian Footballers. **Methods:** Twenty-five male athletes undertook NHE (n = 13) or ISO (n = 12) training across a 38-week period (including pre-season and in-season). Biceps femoris long head (BFlh) architecture, isometric and eccentric knee flexor strength were assessed at baseline, end of pre-season (14-weeks) and at the conclusion of the intervention. Sprint times and force-velocity profiles were determined at baseline and at the end of pre-season. **Results:** Following the intervention both groups had significant improvements in BFlh fascicle length (NHE: 1.16cm, 95%CI 0.68 to 1.63cm, *d*=1.88, p<0.001; ISO: 0.82cm, 95%CI 0.57 to 1.06cm, *d*=1.70, p<0.001), muscle thickness (NHE: 0.11cm, 95%CI 0.01 to 0.21cm, *d*=0.51, p=0.032; ISO: 0.21cm, 95%CI 0.10 to 0.32cm, *d*=0.86, p=0.002) and eccentric strength (NHE: 83N, 95%CI 53 to 114N, *d*=1.79, p<0.001; ISO: 83N, 95%CI 17 to 151N, *d*=1.17, p=0.018). Both groups also finished the intervention weaker isometrically than they started (NHE: -45N, 95%CI -81 to -8N, *d*=-1.03, p=0.022; ISO: -80N, 95%CI -104 to -56N, *d*=-3.35, p<0.001). At the end of pre-season the NHE group had improved their 5m sprint time by 3.3% (±2.0%) and their maximum horizontal velocity was 3% (±2.1) greater than the ISO group who saw no changes. **Conclusion**: Both ISO and NHE training with a periodised sprinting program can increase BFlh fascicle length, thickness, and eccentric strength in Australian Footballers. NHE training also improves 5m sprint time and maximum velocity. However, both interventions reduced isometric strength. These findings provide unique, contextually relevant insights into the adaptations possible in semi-professional athletes.

Key Words: SKELETAL MUSCLE INJURY AND REPAIR, INJURY PREVENTION

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INTRODUCTION

Hamstring strain injuries (HSIs) remain the most common cause of time lost from competition in field-based team sports (1), such as Australian Rules Football (ARF) (2). Athletes are required to perform regular sprinting efforts during ARF competition (3), which likely elevates the risk of a HSI occurring, due to the biomechanical demands placed on the hamstrings (4). Rather than stopping athletes from sprinting, researchers and practitioners aim to identify interventions that address modifiable variables associated with HSI risk, such as hamstring muscle architecture and strength (5-8). It is important to consider the impact that these interventions have on sprint acceleration performance, given this is a key requirement of ARF and many other field-based team sports.

Including the Nordic hamstring exercise (NHE) as part of a training program reduces HSI risk in athletes (9, 10). This risk reductive effect may be driven by increases in biceps femoris long head (BFlh) fascicle length and eccentric knee flexor strength noted after NHE interventions (11-13). The NHE has also been shown to result in improvements in sprinting performance (11, 14). Despite these apparent benefits, the NHE is still not widely implemented in field-based team sports (15). The lack of NHE implementation may be due to fear of muscle soreness caused by this relatively high-intensity eccentric stimulus (16), which has led to interest in isometric hamstring exercises interventions.

Despite interest in isometric hamstring exercises, there has been limited research into the potential benefits of these interventions on HSI risk factors and sprint performance. One study found that an isometric hip extension exercise leads to greater improvements in the

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single-leg hamstring bridge to fatigue test compared to the NHE (17). However, the single-leg hamstring bridge to fatigue test is not a valid measure of hamstring strength (18), and this study did not investigate sprinting performance or muscle architecture. In a more recent study (19), an isometric hip extension exercise was shown to improve isometric hamstring strength but failed to significantly change biceps femoris long head fascicle length. However, this study did not assess sprinting performance and participants were recreationally active, which highlights the need for further research into the impact of isometric hamstring exercise interventions on HSI risk factors and sprinting performance in athletes.

The aim of this study was to compare the impact of an isometric knee flexion exercise intervention to a NHE intervention on hamstring muscle architecture, strength and sprinting performance in semi-professional ARF athletes across a competitive season.

METHODS

Participants

This interventional cohort trial was undertaken across the pre-season and in-season period of the Victorian Football League (VFL) competition. The VFL is a second tier, semi-professional ARF competition. Ethical approval for the study was granted by the Australian Catholic University Human Research Ethics Committee (approval number: 2019-129E). All athletes from a single VFL team were invited to participate and provide written, informed consent. In total, 25 male athletes (age = 22±3yrs; height = 1.85±0.07m; body mass = 81.3±6.7kgs) were recruited and consented to participating.

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Study design

Prior to the intervention commencing, all participants undertook a familiarisation session with the NHE (Figure 1, Supplemental Video 1, Supplemental Digital Content 1), the isometric knee flexion hold (ISO – Figure 1, Supplemental Video 2, Supplemental Digital Content 2) and all testing methods. Following the familiarisation session (median = 5, range 5-7 days), all participants had the following baseline measures assessed:

1. BFlh architecture;
2. Maximal eccentric knee flexor strength during the NHE;
3. Maximal isometric knee flexor strength with 90° of hip and knee flexion (90/90) (20);
4. 40-metre sprint performance with splits at 5, 10, 20, 30 and 40-metres.
    a. With force-velocity profiles subsequently determined (21).

After these baseline assessments, participants were allocated into either the NHE group (n = 13) or the ISO group (n = 12). Participants in each intervention group only performed the one hamstring-specific resistance training exercise for the entire pre-season and in-season period. Additionally, all participants performed periodised sprinting as part of their on-field football specific training sessions, with the prescribed intensity and volume matched between the groups (Figure 2, Supplemental Tables 1a-1c, Supplemental Digital Content 3, The actual versus planned weekly sprint distance (in metres), sprint efforts, and velocities undertaken at ≥25km/hr for the sprinting intervention, and The actual versus planned weekly sprint efforts undertaken at ≥25km/hr for the sprinting intervention). All resistance and sprint training sessions were undertaken with supervision from the members of research team, as part of their roles in the strength and conditioning department of the football club (DF, VN, JG, NMo). All participants undertook an initial six weeks of training after which there was a four week break for the Christmas holiday period. Over these four weeks the participants were

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specifically instructed to refrain from resistance and sprint training. Following the holiday period there was an additional eight weeks of training in the pre-season period (November to March – total of 14 weeks – Figure 2). The subsequent in-season training period occurred over the remaining 24 weeks between March and August.

Architectural assessments of the BFlh using two-dimensional ultrasound were completed at baseline and after 14 (end of pre-season) weeks as well as at the end of the intervention (38 weeks). Both strength assessments (NHE and 90/90) were undertaken at baseline, after 14 (end of pre-season) and 29 weeks as well as at the end of the intervention (38 weeks). The maximal sprinting assessment was undertaken at baseline and after 14 weeks of training (end of pre-season). During all on-field sessions (sprint training, football specific and competitive matches), global positioning (GPS – OptimEye S5, 10Hz, Catapult Sports, Melbourne, Australia) data were used to monitor the external load metrics of each group. These metrics included: total distance covered, efforts undertaken at >25km/hr, distance covered at >25km/hr, peak and average velocity of sprint efforts which were made relative to each participant's maximum recorded value.

Training intervention

As part of their weekly team resistance training sessions, all participants completed one of either the NHE or ISO hold exercises. During the training intervention, all participants also undertook a periodised sprint training intervention (Figure 2) prior to their on-field football specific sessions. The frequency of both the sprinting and resistance training sessions during the in-season period depended on the match schedule, with either one or two sessions undertaken per week, with the sprinting and resistance training interventions matched between the groups. When two sessions per week (both sprinting and resistance training)

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were undertaken, these were completed on non-consecutive days of the week (typically a Tuesday and Thursday with a Saturday match). The training prescription was matched for the NHE and ISO hold exercises across the entire intervention period and was two sets of four repetitions (Table 1). Both the NHE and ISO hold exercise were undertaken for four second repetitions to match total time under tension between the groups.

Nordic hamstring exercise training. A video of the NHE can be found in Supplemental Video 1 (Supplemental Digital Content 1). Participants undertook the NHE with their ankles secured underneath a fixed piece of gym equipment that allowed a standardised kneeling height and position for all repetitions. This was preferred to the partner held version of the NHE, where the strength of the partner may limit the effectiveness of the effort. For each repetition, participants only performed the lowering (eccentric) portion of the NHE over a four second period. After each repetition, participants were instructed to use their arms and flex their hips to assist themselves in returning to the starting position, avoiding any concentric contraction of the hamstrings. Once additional weight was utilised, the participants were required to hold weight plates totalling 5, 7.5 or 10kg over their xiphoid process, per previous work (REF). All NHE training was preceded by a warm up set of three submaximal, bodyweight repetitions and following each training set participants were afforded a 2 min rest. For all NHE training sessions across the intervention period, participants completed two sets of four repetitions, which has previously been shown as an effective dose to maintain eccentric knee flexor strength and BFlh fascicle lengths in recreational individuals (13).

Isometric hamstring training. A video of the ISO hold can be found in Supplemental Video 2 (Supplemental Digital Content 2). Participants started in a kneeling position, with their ankles secured under a piece of gym equipment. The position was similar to the typical break

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point of the NHE, where their hips were slightly flexed, with a forward lean of approximately 45 degrees at the knee and using their arms to brace the upper body on a bench placed in front of them. With the cues “hips under, quiet bum” participants were instructed to “pull up” as hard and fast as they can against the fixed gym equipment for four seconds. Prior to each training session participants undertook a sub-maximal warm up consisting of three ISO hold repetitions at 50%, 75% and approximately 95% of their perceived maximum. Following each training effort, participants were afforded a 30 second rest prior to the next attempt, with two minutes rest between sets. For all ISO training sessions across the intervention period, participants completed two sets of four repetitions, with training time matched between the NHE and ISO groups.

Periodised sprint training. Prior to all on-field football specific training sessions, participants undertook a periodised sprint intervention for the entire study period (Supplemental Tables 1a – 1c, Supplemental Digital Content 3). All sprint training was preceded by an on-field group warm up, following which participants undertook sub-maximal sprinting efforts prior to any maximal attempts. For all the maximal efforts, participants were paired together based on their maximal sprint velocity and undertook their efforts in a race format where they were motivated to run faster than the person next to them. Across the study period, the number and distance of the sprinting exposures was periodised to provide a progressive stimulus (Figure 2). This periodisation was planned on the assumption that the participants could potentially not undertake any sprinting efforts during their football specific sessions or matches. Therefore, should this occur, they will have still been exposed to a sprinting stimulus as part of this prescription. During all sprint training efforts (≥25km/hr), velocity was collected utilising GPS data (OptimEye S5, 10Hz, Catapult Sports, Melbourne, Australia) and recorded as absolute and relative values. Relative velocity was determined as a percentage of each

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participant's maximum recorded velocity, which was updated if it increased throughout the study.

Outcome measures

BFlh architecture. The methods to assess BFlh architecture has been previously reported (5, 22, 23) Briefly, muscle thickness, pennation angle and fascicle length of the BFlh was determined from ultrasound images taken along the longitudinal axis of the muscle belly utilising a two-dimensional, B-mode ultrasound (frequency, 12Mhz; depth, 8cm; field of view, 14 x 47mm) (GE Healthcare Vivid-*i*, Wauwatosa, U.S.A). The scanning site was determined as the halfway point between the ischial tuberosity and the knee joint fold, along the line of the BFlh. All architectural assessments were performed with the participant prone on a massage plinth, after 5mins of inactivity. The orientation of the probe was then manipulated by the assessor (RGT) whose reliability has been previously reported (23).

Once the images were collected, analysis was undertaken off-line (MicroDicom, Version 0.7.8, Bulgaria). Muscle thickness was defined as the distance between the superficial and intermediate aponeuroses of the BFlh. Pennation angle was defined as the angle between the inferior aponeurosis and a fascicle of interest. The aponeurosis angle for both aponeuroses was determined as the angle between the line marked as the aponeurosis and an intersecting horizontal reference line across the captured image (24, 25). As the entire fascicle was not visible in the field of view of the probe, its length was estimated via the following equation (25):

$$FL = \sin(AA + 90°) \times MT / \sin(180° - (AA + 180° - PA)).$$

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Where FL=fascicle length, AA=aponeurosis angle, MT=muscle thickness and PA=pennation angle. Fascicle length was reported in absolute terms (cm). The extrapolation measure and equation has been validated against cadaveric BFlh tissue and is considered a robust way of estimating fascicle lengths (25).The same assessor (RGT) collected and analysed all scans and was blinded to participant identifiers during the analysis.

Eccentric knee flexor strength. The assessment of eccentric knee flexor strength during the NHE has previously been reported (26, 27). Participants knelt on the device (NordBord, Vald Performance, Queensland, Australia) while an investigator placed ankle braces superior to each lateral malleolus. Whilst in this kneeling position, participants were required to cross their arms over their chest or position a weight plate (as required) over the xyphoid process. Once the movement commenced, participants were cued to keep their hips in a position of full extension throughout the effort. Only the eccentric phase of the NHE was completed. Participants undertook a standardised warm up protocol consisting of one repetition at each of 50, 75 and 95% of their perceived maximal effort at bodyweight. Following this, participants completed one set of three maximal NHE repetitions. The average of the three peak values (in Newtons (N)) during the assessment was used for the analysis.

Isometric knee flexor strength. The assessment of maximal isometric knee flexor strength used in this study has previously been reported (20), showing excellent levels of reliability (coefficient of variation = <5.5%; intraclass correlation = 0.95). All assessments were undertaken with the force plates (ForceDecks FD4000, Vald Performance, Queensland, Australia, sampling frequency: 1000Hz) positioned on top of a wooden box. Participants were required to lay supine, with the heel of the limb being tested positioned on top of the

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force plate with the hip and knee both positioned at 90° of flexion. The non-testing limb was rested on the floor next to the wooden box and was held in place by one of the investigators. Participants were instructed to push their heel into the force plate as hard and fast as possible without lifting their hips or head off the floor. Participants undertook a standardised warm up protocol consisting of one repetition at each of 50, 75 and 95% of their perceived maximal effort. Following this, participants completed one set of two maximal isometric knee flexion repetitions. Each maximal effort was undertaken over a period of three seconds with a 30 second rest between contractions. The peak force value (in Newtons (N)) during the assessment was used for the analysis.

Sprint assessment. Sprint performance was assessed using a 40-m overground sprint, with timing gates (SmartSpeed, Fusion Sport, Brisbane, Australia) collecting split times at 5, 10, 20, 30 and 40-m. Participants initiated the sprint themselves by maximally accelerating from a staggered standing start in line with the first timing gate. Each gate was set at a height approximate to the level of the hip when in upright stance. Each participant completed two trials, with the best 40m attempt used for all analysis.

A custom-made Microsoft Excel spreadsheet derived from the work of Samozino et al. (21) was used to determine the theoretical maximal horizontal force (F_0) and velocity (V_0) using the data obtained during the sprint assessment. These values were determined using the quickest sprint test, its associated split times and the body mass of the participant. Using the spreadsheet, any calculations where the squared differences (e.g., the difference between the experimental and estimated values) was >0.1, were excluded from further analyses. The maximal horizontal force and velocity outputs were reported as N/kg and metres per second (m/s), respectively.

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Statistical analysis

All statistical analyses were performed using the R statistical programming language and the following packages: dplyr, lme4 and car (28). Where appropriate, data were screened for normal distribution using the Shapiro-Wilk test. Linear mixed models were fitted and used to assess changes in each of the outcome variables across the study period as well as comparing the sprinting loads between groups. For each outcome variable, covariates were group (NHE or ISO) and time, with participant ID included as a random effect. For BFlh architecture, eccentric and isometric knee flexor strength, the left and right limbs were averaged as they did not differ at any time point. Where significant main or interaction effects were detected, *post hoc* t-tests were used to determine where any differences occurred. Significance was set at $p<0.05$ and where possible Cohen's *d* was reported for the effect size of the comparisons, with the levels of effect being deemed small (d = 0.20 to 0.49), medium (d = 0.50 to 0.79) or large ($d \geq 0.80$). All data were expressed as mean ± SD, unless otherwise stated.

Power calculation

Power analysis was undertaken *a priori* using G-Power (29). The effect size estimates used in the analysis were based on the estimated changes in BFlh fascicle length following a similar NHE training intervention (30). In this previous study, fascicle length changes had an effect size of 2.8. Considering we utilised a lower volume and intensity prescription of the NHE in this intervention, a conservative effect size of 1.4 was deemed a reasonable starting point. Power was set at 95%, with an alpha of 0.05 returning a calculated minimum sample of 12 per group.

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RESULTS

Participant details and intervention compliance

The two groups were similar with respect to height and body mass (NHE height 183±8cm, body mass 79.5±7.2kg; ISO height 185±6cm, body mass 83.4±5.6kg; height p=0.445, body mass p=0.152). However, on average the ISO group (24±3 years) were older than the NHE cohort (21±2 years; p=0.02). Of the 76 total resistance training sessions, the NHE group had an average compliance of 81±15% (range: 41 to 76 sessions completed). The ISO group had a similar level of compliance with 78±14% (range: 40 to 76 sessions completed).

External load metrics

There were no significant differences between the groups, at any of the weeks throughout the intervention when comparing the number of weekly sprinting efforts (p range across all weeks = 0.231 to 0.982) and distance covered ≥25km/hr (p range across all weeks = 0.203 to 0.986), as well as the average relative velocity of these sprinting efforts ≥25km/hr (p range across all weeks = 0.161 to 0.998).

BFlh architectural characteristics

A summary of the BFlh architectural variables during this intervention can be found in Figure 3 and Supplemental Table 2a (Supplemental Digital Content 3).

Fascicle length. There was a significant main effect for time (p<0.001), but not for group (p=0.081) on BFlh fascicle length. Both the NHE and ISO groups had a significant increase

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in fascicle length from baseline to the end of pre-season (NHE group: difference 0.73cm, 95% CI 0.53 to 0.94cm, d = 1.33, p < 0.001; ISO group: difference 0.51cm, 95% CI 0.34 to 0.67cm, d = 0.92, p < 0.001). At the end of the intervention, both groups saw a further significant increase in fascicle length compared to the end of pre-season (NHE group: difference 0.42cm, 95% CI 0.03 to 0.88cm, d = 0.68, p = 0.038; ISO group: difference 0.31cm, 95% CI 0.05 to 0.57cm, d = 0.59, p = 0.021). These changes were also significantly increased when compared to baseline measures for both groups (NHE group: difference 1.15cm, 95% CI 0.68 to 1.63cm, d = 1.88, p < 0.001; ISO group: difference 0.81cm, 95% CI 0.58 to 1.06cm, d = 0.81, p < 0.001). The NHE group had significantly shorter fascicles than the ISO group at baseline (difference 0.57cm, 95% CI 0.13 to 1.01cm, d = 1.07, p = 0.013), but there were no differences at the other time points (d range: 0.39 to 0.59, p range: 0.151 to 0.322).

Muscle thickness. BFlh muscle thickness showed a significant main effect for group (p=0.013) and time (p<0.001). Post hoc analysis indicated that both groups demonstrated a significant increase in muscle thickness from baseline to the end of the intervention (NHE: difference 0.11cm, 95% CI 0.01 to 0.21cm, d = 0.51, p = 0.032; ISO: difference 0.21cm, 95% CI 0.10 to 0.32cm, d = 0.86, p = 0.002). However, only the ISO group demonstrated a significant increase when compared to the end of pre-season (difference 0.13cm, 95% CI 0.01 to 0.25cm, d = 0.52, p = 0.048), whereas the NHE group did not (difference 0.03cm, 95% CI -0.03 to 0.11cm, d = 0.19, p = 0.332). Between-group comparisons showed that the ISO group had a significantly thicker muscle than the NHE group at the end of the intervention (difference 0.26cm, 95% CI 0.10 to 0.42cm, d = 1.37, p = 0.002), but there were no differences at other time points.

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Pennation angle. There were no significant main effects for group (p=0.188) or time (p=0.471) found for BFlh pennation angle. Therefore, no *post hoc* analyses were undertaken.

Eccentric knee flexor strength

A significant main effect for time (p<0.001), but not group (p=0.503) was found for eccentric knee flexor strength. *Post hoc* analyses showed a significant increase in eccentric knee flexor strength at the end of the intervention when compared to the end of pre-season for both groups (NHE: 83N, 95% CI 53 to 114N, *d* = 1.79, p < 0.001; ISO: 83N, 95% CI 17 to 151N, *d* = 1.17, p = 0.018; Figure 4 and Supplemental Table 2b, Supplemental Digital Content 3). There were no differences between groups at any time point (*d* range = 0.02 to 0.35, p range = 0.384 to 0.940).

Isometric knee flexor strength

A significant main effect for time (p<0.001), but not group (p=0.340) was found for isometric knee flexor strength. Post hoc analyses showed a significant decrease in isometric knee flexor strength from baseline to the end of pre-season for both groups (NHE: -28N, 95%CI -51 to -5N, d = -0.58, p = 0.020; ISO: -39N, 95% CI -65 to -13N, d = -1.10, p < 0.001; Figure 4 and Supplemental Table 2b, Supplemental Digital Content 3). These reductions persisted through the intervention for both groups (NHE: -45N, 95% CI -81 to -8N, d = -1.03, p = 0.022; ISO: -80N, 95% CI -104 to -56N, d = -3.35, p < 0.001), with only the ISO group getting weaker during the in-season period (difference in NHE: -16N, 95% CI -46 to 16N, d = -0.47, p = 0.268; difference in ISO: -41N, 95% CI -74 to -8N, d = -1.18, p = 0.019). The NHE group was stronger isometrically than the ISO group at the end of the intervention (difference 29N, 95% CI 9 to 49N, d = 1.20, p = 0.006).

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Sprint assessment

Split results. A significant main effect for time (p=0.003), but not group (p=0.619) was found for 5m split times. *Post hoc* analyses showed a significant decrease in 5m split times (i.e., improved performance) from baseline to the end of the pre-season period in the NHE group only (NHE: -0.037s, 95% CI -0.681 to -0.014s, *d* = -0.83, p = 0.043; ISO: -0.021N, 95% CI -0.693 to 0.010, *d* = -0.82, p = 0.056; Figure 5 and Supplemental Table 2c, Supplemental Digital Content 3). There were no differences in 5m split times at any time point when comparing between the groups (d range = -0.07 to 0.72 and p range = 0.102 to 0.876).

A significant main effect for group (p=0.009) and time (p=0.023) was also found for 40m sprint performance. *Post hoc* analyses found that the NHE group was significantly faster over 40m when compared to the ISO group at the end of pre-season (Difference 0.110s, 95% CI 0.006 to 0.214s, *d* = 0.87, p = 0.039). There were no other significant within or between group changes in 40m sprint performance (*d* range = 0.03 to 0.72, p range = 0.085 to 0.993).

Horizontal force-velocity profile. A significant main effect for group (p=0.007) and time (p=0.021) was found for maximal horizontal velocity. *Post hoc* analyses found that the NHE group had a significantly greater horizontal velocity than the ISO group at the end of pre-season (Difference 0.26m/s, 95% CI 0.01 to 0.51m/s, *d* = 0.82, p = 0.049; Supplemental Table 3, Supplemental Digital Content 3). There were no other significant within or between group changes in horizontal force-velocity profiles (*d* range = 0.06 to 0.82, p range = 0.057 to 0.876)

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DISCUSSION

This is the first study to investigate the impact of a long-term (38 week) NHE or isometric knee flexor training intervention, alongside a periodised sprinting program, on BFlh architecture, strength and sprinting performance in ARF athletes. The novel findings of this study are:

1. Undertaking a NHE or isometric knee flexor training intervention, alongside a periodised sprinting program, significantly improves BFlh fascicle length and muscle thickness across 38 weeks.
2. Both interventions also significantly increase eccentric strength across the in-season period, with no alterations during pre-season.
3. Isometric knee flexor strength significantly reduced at the end of both the pre-season for both groups, with the ISO group getting weaker in-season.
4. Undertaking NHE training, alongside a periodised sprinting program, across a 14 week pre-season period can significantly improve 5m split time, and
5. A NHE intervention, alongside a periodised sprinting program, can significantly improve maximal sprinting velocity over 40m when compared to an ISO training intervention after a 14 week pre-season period.

This is the first study to prescribe a single-mode hamstring specific exercise intervention, alongside a periodised sprinting program, in semi-professional competitive athletes. Previous research in this area is limited with its application to athletic populations as it has either implemented a single exercise intervention in recreationally active participants (12, 13, 19, 31) or has not controlled and prescribed a periodised sprinting program as part of the training prescription (11). The key inclusion of the periodised sprinting program alongside a single-

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mode hamstring specific exercise intervention in this study allows for a more holistic and translatable set of findings as it shows the outcomes when both are prescribed together, which is commonplace in applied settings.

Elite ARF athletes who possess short BFlh fascicles are more likely to suffer a HSI than those who have longer fascicles (7). In a similar elite ARF cohort, possessing low levels of eccentric knee flexor strength has also been shown to increase the likelihood of a HSI occurring (32), although this is not a consistent finding (6, 7, 33). Whilst it is unknown if modifications in these variables are associated with a change in the risk of HSI occurring, those who undertook the NHE intervention in the current study saw a 1.15cm (11%) and a 38N (9%) increase in fascicle length and eccentric strength across the study, respectively. Similarly, the ISO training group saw a 0.82cm (7%) and a 49N (12%) increase in both fascicle length and eccentric strength respectively. The findings for the ISO training group in the current study are the first to show that it is possible to increase BFlh fascicle length and eccentric strength while undertaking a period of isometric training. However, the current study does differ compared to previous isometric hamstring training studies (17, 19). All previous interventions have been undertaken using a single-exercise approach (17, 19), whereas the current study includes a periodised sprinting program which was undertaken alongside the resistance training stimuli. As the BFlh is hypothesized to undergo significant lengthening during sprinting (4), it is possible that the inclusion of the periodised sprinting program may have contributed to the increases in BFlh fascicle length and eccentric strength for both groups. There is also some evidence to support the idea that sprint training alone is capable of increasing BFlh fascicle lengths (34). When considering these findings alongside the prospective risk factor evidence for BFlh fascicle length and eccentric strength, it is possible that completing NHE or ISO exercise alongside a periodised sprinting program may

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have a beneficial effect on HSI risk mitigation, although this would require further investigation.

Muscle thickness of the BFlh was significantly increased after the pre-season period in both groups. This is the first evidence that isometric training interventions can improve muscle thickness in the BFlh (19), whereas NHE interventions have previously been shown to increase muscle thickness (13, 34). Similar to the fascicle length improvements in the ISO group, the inclusion of a sprint training stimulus in the current study may have contributed to these improvements as a period of sprint only training has been shown to increase BFlh muscle thickness (34). These results suggest that training programs aiming at improving muscle thickness and fascicle length of the BFlh could include either an eccentric or isometric resistance training stimuli alongside a periodised sprinting program.

We found a significant reduction in isometric knee flexor strength for both groups at the end of the intervention. An assessment of isometric knee flexor strength was included in the current study to assist in the detection of any contraction mode specific adaptations to the two divergent training interventions. However, there are several factors which may need to be considered when interpreting these findings. It has been previously shown that there is a joint-angle specificity for adaptations following isometric training interventions (35). As such in the ISO group, the difference in joint position between the isometric strength testing (90° hip and knee flexion) and training stimuli (approx. 45° at the knee and a neutral hip) may have led to the reductions seen. The isometric strength testing position used in this study has been previously shown as reliable when implemented in athletic populations (20). Previous evidence has found no association between isometric strength and HSI risk (8), however this

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assessment of isometric knee flexor strength is regularly used as part of screening batteries to track athlete recovery (36) and training management as required (37). Considering the joint-angle specificity of isometric training adaptations (35), it is possible that if we had tested isometric knee flexor strength in the joint positions where it was trained, we may have seen significant improvements in the ISO group. However, the assessment of isometric knee flexor strength was chosen because of its proven reliability and its capacity to allow a greater throughput when assessing individuals in an applied setting. Conversely, undertaking a period of eccentric training has been shown to improve eccentric but not isometric strength (19). Therefore, whilst the testing position may have impacted the ability to see significant increases in the ISO group, in the NHE group it may be a case of a contraction mode specific adaptation leading to the current findings. Considering these findings alongside the methodological intricacies of the current study, future research should consider measuring isometric strength at various joint positions in athletic cohorts to determine if there is a joint-angle specificity to the adaptations seen following a period of isometric training.

In ARF, the evolution of the game has resulted in a significant increase of the number of short-distance sprinting efforts (3). As a result, high performance programs aim to improve acceleration capacity (5m sprint performance) more than longer distance sprinting efforts (e.g., over 30m) (38). In the current study, at the end of pre-season the NHE group had significantly improved their 5m split time by 3.3%. To contextualise these findings in a sporting context, an average participant from the NHE group may gain an extra 15 to 20cm advantage across 5m when competing against an ISO group participant. In addition to this, the NHE group also possessed a greater maximum velocity than the ISO group at the end of the pre-season. In ARF this improvement may be crucial to improving an athlete's ability to

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best their opponent during contests (3). The improvement in a performance metric such as 5m split time may also assist the "buy-in" and implementation of an exercise-based injury prevention strategy in a high-performance setting where the balance between performance and prevention is a key component of the role (39).

There are limitations in the current study. The measurement of fascicle length from the BFlh is an estimation made using an equation validated in comparison to cadaveric data (24, 25). This estimation is required due to the small field of view being unable to capture the entire fascicle. This methodology was chosen due to fascicle length measures using it have been prospectively associated injury (8). This technique has also been shown to be reliable and have a high level of agreement with cadaveric samples (25). Minimal clinically importance difference values for the architectural or strength measures weren't determined as no previous interventions have directly investigated whether changes in fascicle length or eccentric strength are associated with a reduction in injury incidence. Finally, there was no control group and as such we were unable to determine the impact that ARF or sprint training alone would have on the outcome measures. However, this is the first study to prescribe a periodised sprinting program alongside a single-exercise intervention to ensure that as much of the stimuli related to hamstring adaptation were controlled.

CONCLUSIONS

In conclusion, undertaking an NHE or isometric knee flexor training intervention alongside a periodised sprinting program appears to improve fascicle length, muscle thickness and eccentric knee flexor strength in ARF athletes. Whereas isometric knee flexor strength appears to decline with either intervention, although this could be due to joint angle or contraction mode specific adaptations. Additionally, undertaking a NHE training

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intervention, alongside a periodised sprinting program, significantly improves acceleration capacity and maximum velocity. These findings provide unique, contextually relevant insights into the adaptations that may be possible in trained athletes.

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Acknowledgments

The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

Conflict of Interest

A co-author of this paper, A/Prof David Opar is listed as a co-inventor on a patent, filed by the Queensland University of Technology (QUT), for a field-testing device of eccentric hamstring strength, which is now known commercially as the NordBord. A/Prof Opar has received revenue distributions from QUT based on revenue that QUT has generated through the commercialisation of his intellectual property. A/Prof Opar is a minority shareholder in Vald Performance Pty Ltd, the company responsible for commercialization of the NordBord and ForceDecks, among other devices. A/Prof Opar has received research funding from Vald Performance, for work unrelated to the current manuscript. A/Prof Opar was previously the Chair of the Vald Performance Research Committee, a role that was unpaid. A/Prof Opar has family members who are minor shareholders and/or employees of Vald Performance

Co-authors of this paper, Christopher Pollard and Joshua Ruddy, were previously employed with Vald Performance who manufacture the Nordbord and ForceDecks used in this study. However, neither were employed by Vald Performance at the time of completing this study.

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FIGURE LEGENDS

Figure 1: Examples of the exercises prescribed for either the Nordic hamstring exercise group (A Panels) or the Isometric group (B Panel).

Figure 2: An overview of the sprinting distance (A) and efforts (B) prescribed across the intervention period.

Figure 3: Architectural measures of the biceps femoris long head at baseline, end of pre-season and the end of the intervention. A = Fascicle length, B = Muscle thickness, C = Pennation angle. Significant differences ($p<0.05$) within each group as well as between groups are indicated on each panel.

Figure 4: Eccentric (A) and isometric (B) measures of knee flexor strength at baseline, end of pre-season and the end of the intervention. Significant differences ($p<0.05$) within each group are indicated on each panel.

Figure 5: Sprint assessment splits of 5 (A), 10 (B), 20 (C), 30 (D) and 40 metres during testing at baseline and the end of pre-season. Significant differences ($p<0.05$) within each group as well as between groups are indicated on each panel.

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SUPPLEMENTAL DIGITAL CONTENT

SDC 1: Supplemental Content 1.mp4: A video of the Nordic hamstring exercise.

SDC 2: Supplemental Content 2.mp4: A video of the isometric hamstring exercise

SDC 3: Supplemental Digital Content.docx

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Figure 1

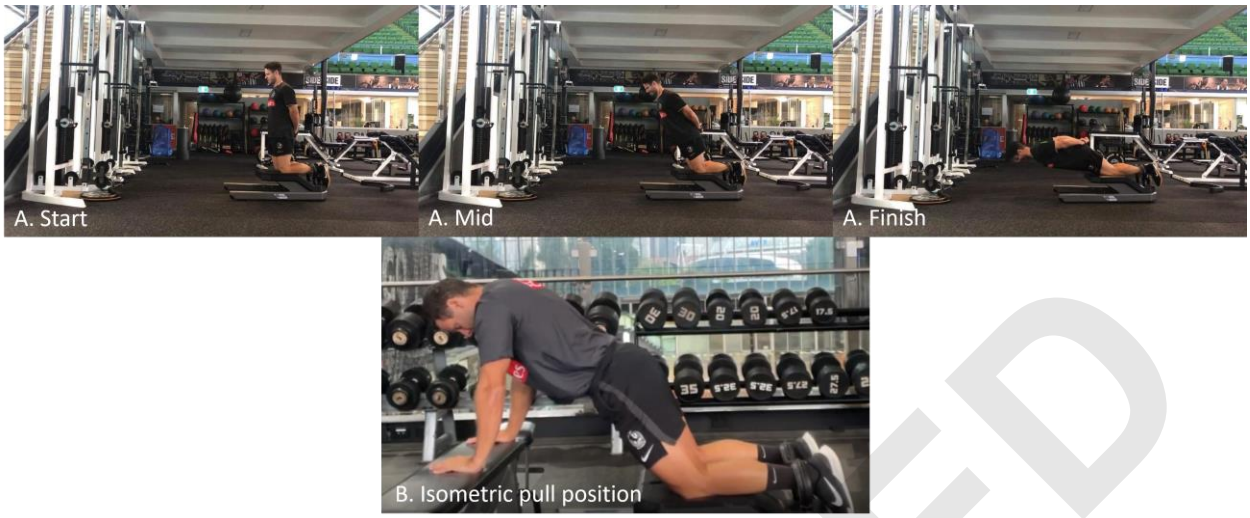

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Figure 2

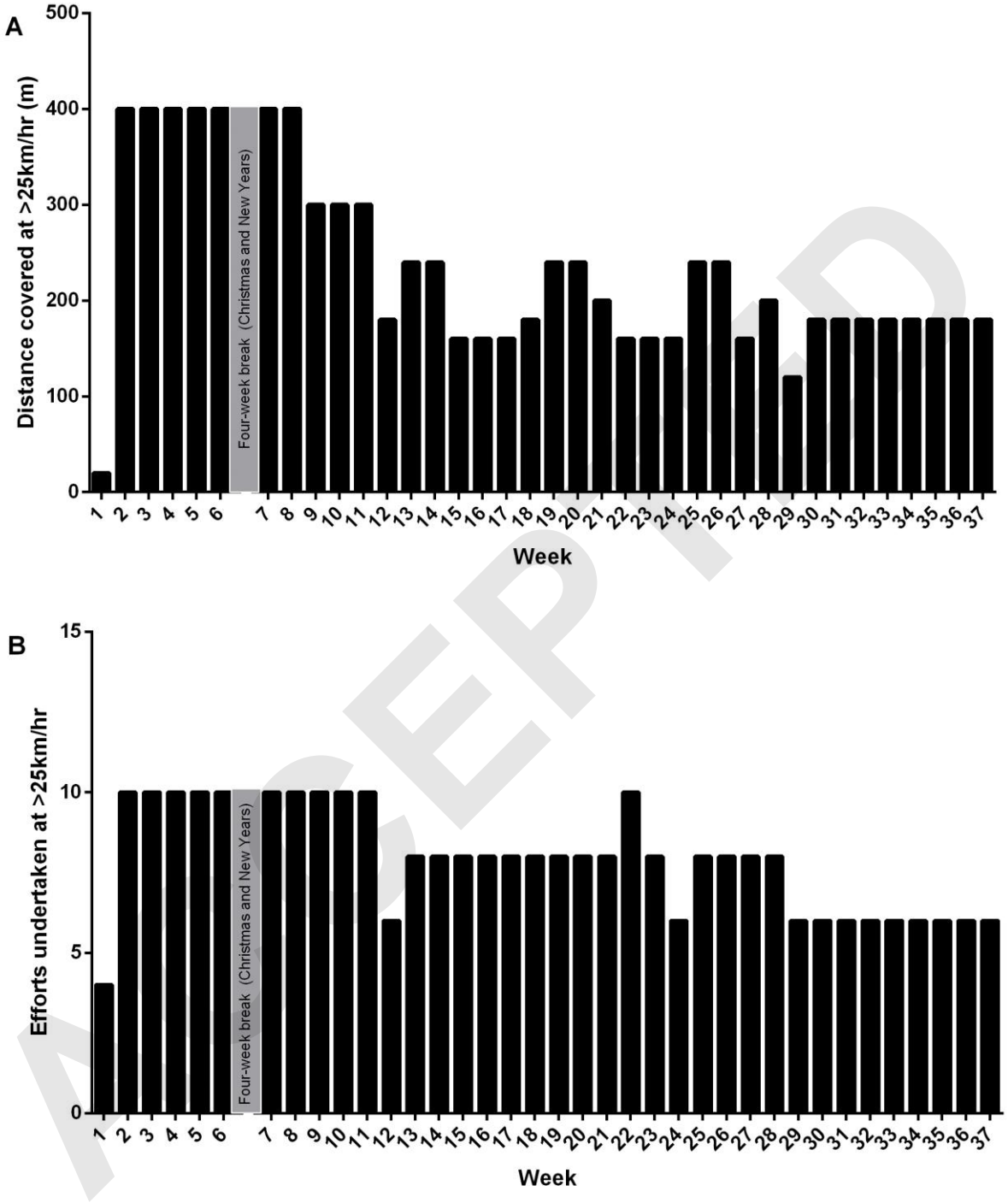

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Figure 3

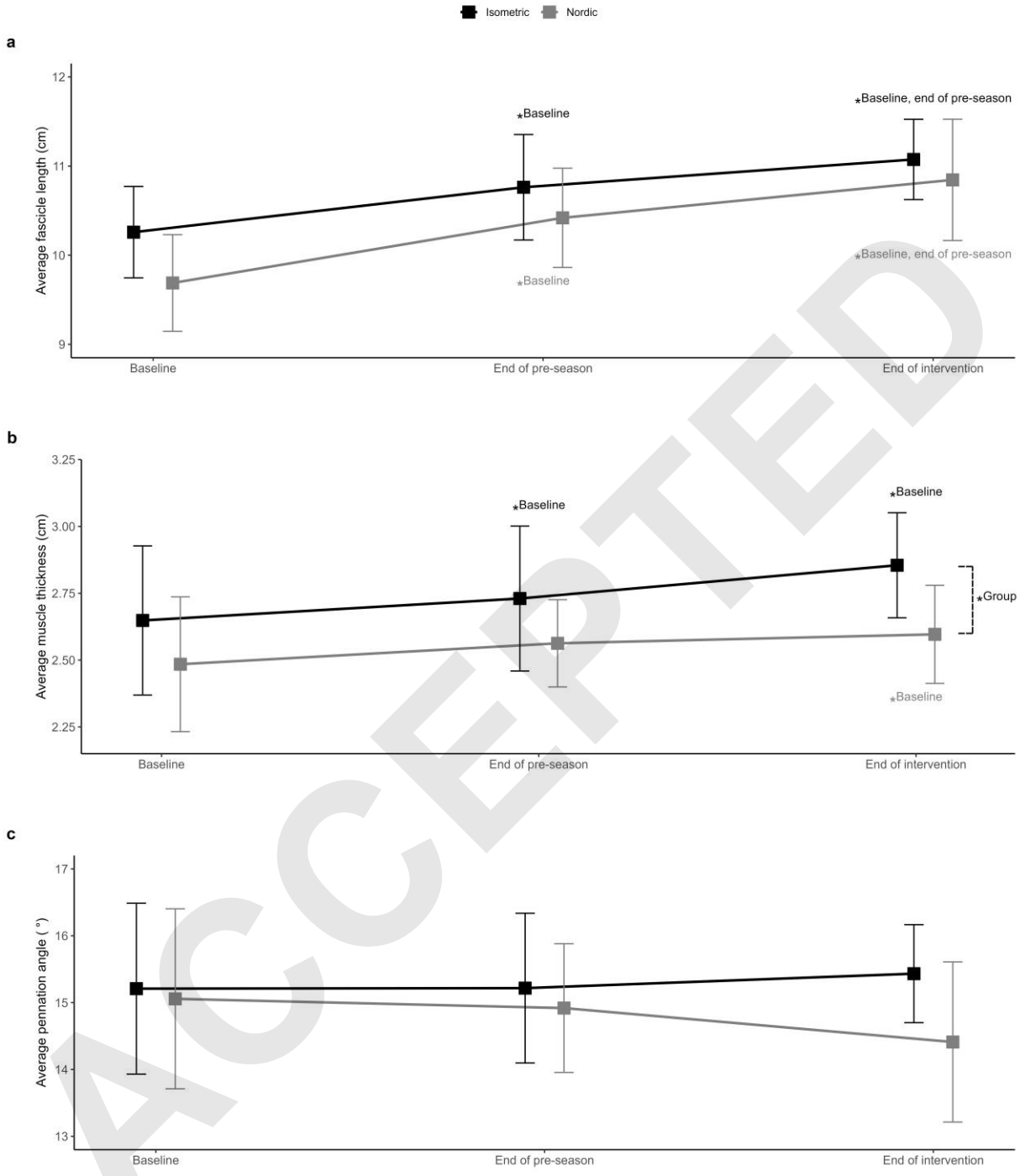

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Figure 4

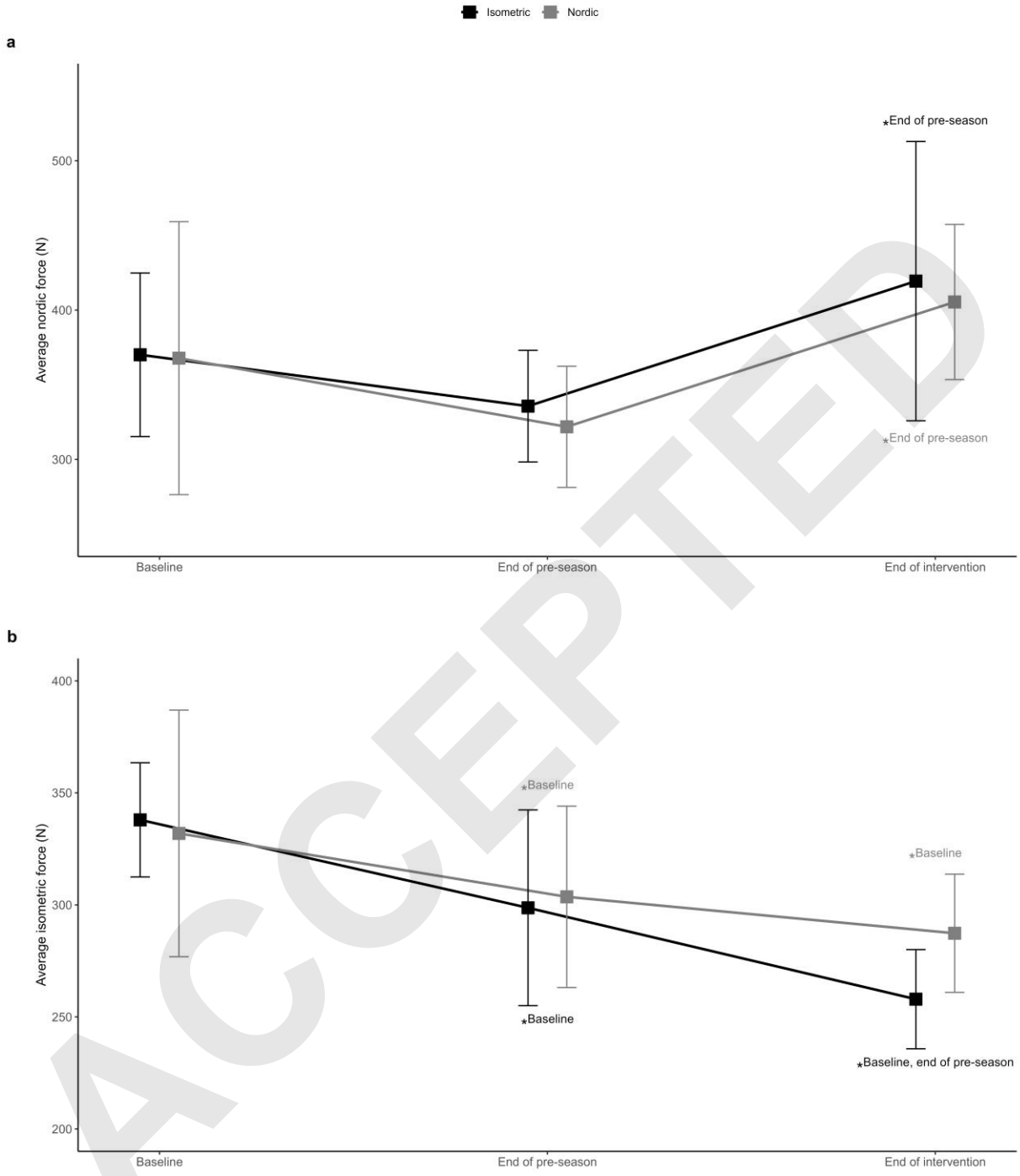

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Figure 5

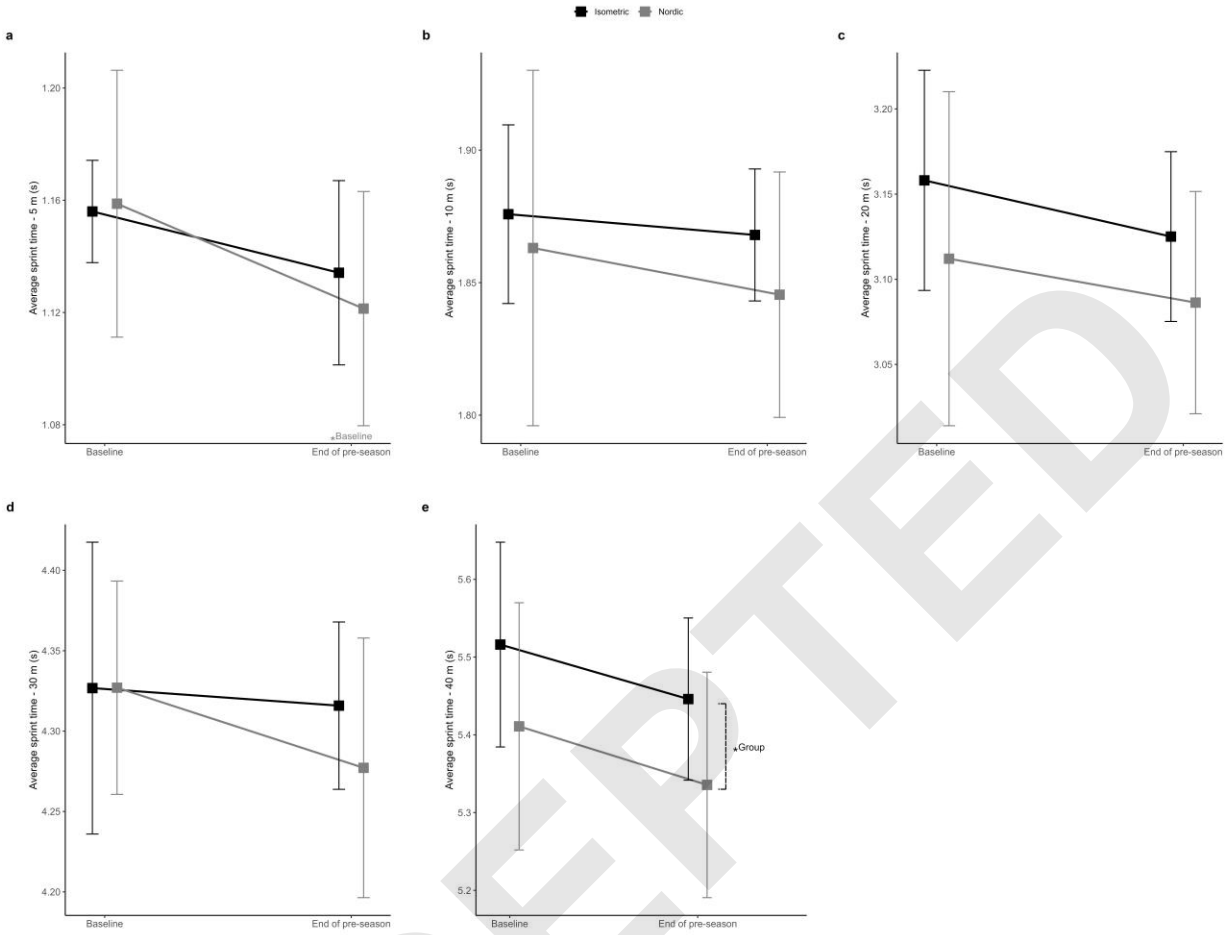

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Table 1 – Prescription of the Nordic and Isometric training efforts for the 37 week training period.

Exercise	Sets	Repetitions	Repetition Time	Inter-set Rest	Resistance
Nordic	2	4	4secs	2mins	Body weight, 5, 7.5, 10kg*
Isometric	2	4	4secs	2mins	Nil[#]

* Additional resistance included for the Nordic Hamstring Exercise in a four week cycle: week 1: body weight, week 2: 5kg, week 3: 7.5kg, week 4: 10kg. After each cycle, participants returned to bodyweight efforts and began the cycle again. This process was followed for the entire length of the intervention period (37 weeks, 9 four week cycles).

[#] All isometric repetitions were performed with maximal effort.

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Supplemental Table 1a: Actual versus planned weekly sprint distance (in metres) undertaken at ≥25km/hr for the sprinting intervention.

On average, the actual weekly sprint distance covered at ≥25km/hr (322.7±301.9m) was greater than what was planned as part of the sprinting intervention (233.0±88.3m) (difference = 89.7m; 95%CI 68.6 to 110.8, *d*=0.40, p<0.001).

Week	Actual (m)	Planned (m)	Difference (m)
1	43.6 ± 14.3	20	23.6
2	586.3 ± 89.8	400	186.3
3	789.2 ± 127.3	400	389*
4	750.0 ± 116.5	400	350*
5	191.9 ± 20.2	400	-208.1**
6	676.9 ± 100.5	400	276.9*
Four week break (Christmas and New Years)			
7	602.6 ± 85.9	400	202.6*
8	1086.9 ± 148.5	400	686.9**
9	269.8 ± 27.7	300	-30.2
10	228.7 ± 13.7	300	-71.3**
11	299.8 ± 36.9	300	-0.2
12	138.7 ± 22.3	180	-41.3
13	428.0 ± 57.8	240	188.0*
14	310.6 ± 62.5	240	70.6
End of pre-season period			
15	272.6 ± 40.9	160	112.6*
16	226.3 ± 37.1	160	66.3
17	292.2 ± 35.2	160	132.2*
18	313.4 ± 40.8	180	133.4*
19	265.7 ± 45.9	240	25.7
20	275.1 ± 40.1	240	35.1
21	393.3 ± 36.1	200	193.3**
22	137.8 ± 25.6	160	-22.2
23	250.3 ± 33.9	160	90.3*
24	308.6 ± 32.0	160	148.6**
25	275.7 ± 32.9	240	35.7
26	219.9 ± 23.3	240	-20.1
27	121.9 ± 16.8	160	-38.1
28	324.0 ± 43.0	200	124.0*
29	184.0 ± 25.8	120	64.0*
30	262.2 ± 45.6	180	82.2*
31	195.8 ± 38.8	180	15.8
32	179.9 ± 21.6	180	-0.1
33	164.0 ± 25.3	180	-16.0
34	188.3 ± 23.6	180	8.3
35	208.1 ± 24.0	180	28.1
36	199.0 ± 22.0	180	19.0
37	215.7 ± 21.8	180	35.7

*= p<0.05, **= p<0.001 when comparing the actual to the planned variable.

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Supplemental Table 1b: Actual versus planned weekly sprint efforts undertaken at ≥25km/hr for the sprinting intervention.

On average the actual weekly sprint efforts undertaken (11.5±8.1 efforts per week) was significantly greater than what was planned (8.0±1.5 efforts per week) as part of the intervention (difference = 3.5 efforts per week; 95%CI 2.9 to 4.1, *d*=0.60, p<0.001).

Week	Actual	Planned	Difference
1	2.7 ± 0.8	4	-1.3
2	14.6 ± 2.5	10	4.6
3	21.6 ± 3.1	10	11.6*
4	18.0 ± 2.3	10	8*
5	7.9 ± 0.9	10	-2.1*
6	14.3 ± 1.8	10	4.3*
Four week break (Christmas and New Years)			
7	13.4 ± 1.8	10	3.4*
8	21.1 ± 2.9	10	11.1*
9	10.7 ± 1.2	10	0.7
10	9.9 ± 0.7	10	-0.1
11	12.0 ± 1.8	10	2.0
12	6.1 ± 0.9	8	-1.9
13	18.2 ± 2.2	12	6.2**
14	9.2 ± 1.5	8	1.2
End of pre-season period			
15	12.5 ± 1.8	8	4.5*
16	10.1 ± 1.7	8	2.1
17	13.3 ± 1.6	8	5.3*
18	14.5 ± 2.0	8	6.5*
19	11.0 ± 2.3	8	3
20	10.7 ± 1.5	8	2.7
21	14.5 ± 1.7	8	6.5**
22	8.8 ± 1.6	10	-1.2
23	10.7 ± 1.7	8	2.7
24	11.1 ± 1.2	6	5.1**
25	12.6 ± 1.6	8	4.6*
26	10.2 ± 1.3	8	2.2
27	6.2 ± 0.9	8	-1.8
28	13.4 ± 1.7	8	5.4*
29	8.0 ± 1.2	6	2.0
30	11.2 ± 2.0	6	5.2*
31	10.1 ± 1.7	6	4.1*
32	7.6 ± 1.0	6	1.6
33	8.3 ± 1.3	6	2.3
34	8.2 ± 1.1	6	2.2
35	6.2 ± 1.0	6	0.2
36	8.1 ± 1.3	6	2.1
37	7.5 ± 1.3	6	1.5

*= p<0.05, **= p<0.001 when comparing the actual to the planned variable

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Supplemental Table 1c: Actual versus planned weekly sprinting velocities relative to each participants absolute maximum velocity across the sprinting intervention.

The average weekly relative speed during these efforts (90.9±6.9%) was less than what was planned (95.0±2.8%) (difference = -4.1%; 95% CI -4.5 to -3.5, *d*=-0.77, p<0.001). The average weekly absolute speed of these sprinting efforts was 31.8±2.2km/hr (range = 25.6 to 36.3km/hr).

Week	Actual (%)	Planned (%)	Difference (%)
1	71.9 ± 1.9	70	1.9
2	92.2 ± 1.3	95	-2.8*
3	92.6 ± 0.6	95	-2.4**
4	91.6 ± 3.1	95	-3.3*
5	93.9 ± 1.8	95	-1.1
6	88.9 ± 1.3	95	-6.1**
Four week break (Christmas and New Years)			
7	91.4 ± 1.6	95	-3.6*
8	91.1 ± 0.9	95	-3.9*
9	93.7 ± 0.6	95	-1.3
10	91.4 ± 1.5	95	-3.6
11	88.4 ± 1.4	95	-6.6**
12	85.1 ± 1.6	95	-9.9**
13	94.0 ± 1.4	95	-1.0**
14	86.1 ± 1.5	95	-8.9**
End of pre-season period			
15	87.4 ± 1.2	95	-7.6**
16	86.8 ± 1.1	95	-8.2**
17	93.4 ± 1.5	95	-1.6
18	93.0 ± 1.3	95	-2.0
19	94.3 ± 0.8	95	-0.7
20	95.6 ± 0.9	95	0.6
21	95.9 ± 0.6	95	0.9
22	86.2 ± 1.4	95	-8.8**
23	92.7 ± 1.2	95	-2.3
24	95.1 ± 1.5	95	0.1
25	93.6 ± 0.6	95	-1.4
26	91.0 ± 1.4	95	-4.0*
27	88.7 ± 1.8	95	-6.3*
28	94.1 ± 0.8	95	-0.9
29	92.0 ± 0.9	95	-3.0*
30	90.6 ± 1.0	95	-4.4*
31	89.3 ± 1.4	95	-5.7**
32	90.4 ± 1.7	95	-4.6*
33	86.4 ± 2.6	95	-8.6*
34	91.2 ± 1.3	95	-3.8*
35	91.0 ± 1.1	95	-4.0*
36	93.4 ± 1.3	95	-1.6
37	91.1 ± 1.3	95	-3.9*

*= p<0.05, **= p<0.001 when comparing the actual to the planned variable.

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Supplemental Table 2a: Biceps femoris long head architectural characteristics during the intervention period for both the isometric and Nordic hamstring exercise groups.

	Isometric Group			Nordic Group		
	Muscle Thickness (cm)	Pennation Angle (°)	Fascicle Length (cm)	Muscle Thickness (cm)	Pennation Angle (°)	Fascicle Length (cm)
Baseline	2.65 ± 0.28	15.2 ± 1.2	10.3 ± 0.5	2.48 ± 0.25	15.1 ± 1.3	9.7 ± 0.5+
End pre-season	2.73 ± 0.28	15.2 ± 1.1	10.8 ± 0.6**	2.56 ± 0.16	14.9 ± 0.9	10.4 ± 0.6**
End intervention	2.85 ± 0.19*#	15.4 ± 0.7	11.1 ± 0.4** #	2.60 ± 0.18*+	14.4 ± 1.2	10.8 ± 0.7** #

All architectural variables are presented as the average of both limbs. * = p<0.05 compared to baseline, ** = p<0.001 compared to baseline, # = p<0.05 compared to end of pre-season, + = p<0.05 comparing between groups

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Supplemental Table 2b: Measures of eccentric and isometric knee flexor strength across the intervention for both the isometric and Nordic hamstring exercise groups.

	Isometric Group		Nordic Group	
	Eccentric knee flexor strength (N)	Isometric knee flexor strength (N)	Eccentric knee flexor strength (N)	Isometric knee flexor strength (N)
Baseline	370 ± 54	337 ± 25	367 ± 91	331 ± 55
End pre-season	335 ± 37	298 ± 43*	321 ± 40	303 ± 40*
End intervention	419 ± 93 #	258 ± 22*	405 ± 51 #	287 ± 26 +

Eccentric and isometric knee flexor strength is the average of both limbs. N = Newtons of force, * = $p<0.05$ compared to baseline, # = $p<0.05$ compared to end of pre-season, + = $p<0.05$ when comparing between groups.

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Supplemental Table 2c: Sprinting assessment split times for both the isometric and Nordic hamstring exercise groups at baseline and the end of pre-season.

	Isometric Group			Nordic Group		
	Horizontal force (N/kg)	Maximum Velocity (m/s)	Maximum Power (W/kg)	Horizontal force (N/kg)	Maximum Velocity (m/s)	Maximum Power (W/kg)
Baseline	11.2 ± 0.5	8.55 ± 0.28	23.9 ± 0.8	11.1 ± 1.1	8.75 ± 0.18	24.3 ± 2.5
End pre-season	11.1 ± 0.4	8.67 ± 0.20	24.3 ± 1.0	11.2 ± 0.8	8.93 ± 0.38+	24.9 ± 1.7

* = $p<0.05$ compared to baseline, + = $p<0.05$ between groups

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Supplemental Table 3: Horizontal force-velocity data for the isometric and Nordic hamstring exercise groups at baseline and the end of pre-season.

	Isometric Group					Nordic Group				
	5m time (s)	10m time (s)	20m time (s)	30m time (s)	40m time (s)	5m time (s)	10m time (s)	20m time (s)	30m time (s)	40m time (s)
Baseline	1.156 ± 0.018	1.875 ± 0.033	3.158 ± 0.064	4.326 ± 0.090	5.516 ± 0.131	1.158 ± 0.047	1.863 ± 0.067	3.112 ± 0.098	4.327 ± 0.066	5.411 ± 0.159
End pre-season	1.134 ± 0.032	1.868 ± 0.025	3.125 ± 0.049	4.315 ± 0.052	5.446 ± 0.104	1.121 ± 0.041 *	1.845 ± 0.046	3.086 ± 0.065	4.277 ± 0.080	5.335 ± 0.145 +

N/Kg = Newtons of force relative to body mass, m/s = metres per second, W/kg = Watts of power relative to body mass, + = $p<0.05$ between groups

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